

Single Carrier Frequency Domain Equalizer for Underwater Wireless Communication

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Abstract

Single carrier modulation with frequency domain equalizer is proposed for underwater acoustic channel. A unique training sequence, viewed as cyclic prefix, is inserted into transmitted sequences so that low complexity minimum mean square error (MMSE) frequency domain equalization (FDE) is used for combating the inter-symbol interference in the receivers. Maximum likelihood estimation for robust initial phase error and Doppler shift are derived by using training sequence. The receiver estimates the initial phase error, Doppler shift and underwater acoustic channel block-by-block. Compare to zero padding or cyclic prefix of single carrier modulation, training sequence has the potential to provide more accurate estimation than decision-directed methods and alleviate the time-scale effect caused by significant Doppler shift for wideband acoustic signal. The proposed UWA communication system with single transmitter and single receiver has been tested in practical South China Sea on June 2007. At a distance of 30 km, the data rates of 450 bps and 900 bps are achieved with uncoded bit error rates of 4 out of 8333 (BPSK) and 103 out of 16666 (QPSK).

I. Introduction

Wireless communications underwater are usually established using acoustic waves. High data rate underwater acoustic (UWA) communication is a challenging problem due to the severe intersymbol interference (ISI) caused by time-varying multipath environments [1]. Over the past two decades, many techniques have been proposed to combat the highly complex and lossy UWA channels. Among them, adaptive multichannel decision feedback equalizers (DFE) with an embedded second order digital phase locked loop (DPLL) to simultaneously track carrier

phase as well as mitigate multipath channel distortions, have been employed successfully in a variety of underwater channels [2]. However, multipath delay spread in some shallow waters which is usually on the order of 10-30 ms causes severe inter-symbol interference (ISI) that extends over several tens or hundreds of symbols for high data rate UWA communication [2]. Over these UWA channels, adaptive multichannel equalizers whose coefficients are updated in the time domain, sample-by-sample in general, like the classical LMS and RLS, are quite demanding in terms of computational complexity, algorithm stability, and selection of receiver parameters, such as the equalizer size and the parameters of the DPLL, for each deployment [3], [4]. Several low complexity time domain equalizers have been studied in the last decade, including reduced-complexity multichannel processing [5], sparse equalizer [6], [7], channel estimated based equalizer [8] and passive time reversal equalizer [9], [10].

Instead of relying on time-domain UWA channel equalizer where the filter coefficients are updated in the time domain, sample-by-sample in general, low complexity, low-maintenance orthogonal frequency division multiplexing (OFDM) [4], [11], [12] have also been investigated for channels whose time variations are sufficiently slow relative to the symbol rate. The frequency domain equalization in OFDM is conducted block-by-block. One key shortcoming of OFDM based UWA communications is the difficulty that confronts acoustic transducers (e.g., hydrophones) encounter when OFDM signals exhibit high peak-to-average power ratio (PAPR). Another difficult challenge on OFDM is the preservation of orthogonal subcarriers under relative high Doppler shift at acoustic wavelength.

Frequency domain equalization in an SC system is computationally simpler than corresponding time domain equalization for channels with severe delay spread as OFDM that equalization is performed in frequency domain block-by-block. When combined with FFT operation and the use of a cyclic prefix, single-carrier with frequency

domain equalization (SC-FDE) delivers the same performance and overall complexity as OFDM [13]. Additionally, SC-FDE modulated signals has smaller PAPR and less Doppler sensitive than OFDM. According to these advantages, SC-FDE has drawn a great attention in broadband wireless communication [13] and UWB communication [14]. In UWA communication, SC-FDE has been considered in [15] and [16] recently. Our earlier work focused on simulated based studies in which CP-based SC-FDE was considered. According to [17], training aided frequency domain equalized single carrier provide better synchronization and channel estimation with the same performance as CP-based SC-FDE.

In this paper, we propose training sequence based SC-FDE for UWA communication. Instead of cyclic prefix, training sequence is used for synchronization, channel estimation and tracking, timing offset and Doppler shift estimation and compensation. The processing is performed block-by-block. Maximum likelihood estimation for initial rotation phase and Doppler shift are presented by using training sequence. Linear minimum mean square error(MMSE) frequency domain equalization is employed. The proposed training sequence based SC-FDE communication is verified by one practical experiment conducted in South China Sea, on June, 2007. At the range of 30km, bit rates of 450bps (BPSK) and 900bps (QPSK) are achieved with the un-coded bit error rate (BER) of $10^{-3} \sim 10^{-4}$ for the single input single output (SISO) system over 500 Hz bandwidth.

The rest of this paper is organized as follows. Section II describes training sequence based SC-FDE. Section III presents details of receiver algorithm for practical UWA communication system. Section IV presents sea experimental before the conclusion remarks of Section V.

II. Training sequence based SC-FDE

Training sequence based single carrier (TS-SC), a modified CP-SC, replaces the CP by a known constant data block. This TS CAN be viewed as a CP by the FFT device and be always the same and hence, not include useful data. It does not need more bandwidth than CP-SC, and achieves synchronization and channel estimation by using the training sequence. In TS-SC, we note N and N_p as the size of each block and the length of training sequence. Among them, the useful data size is $N - N_p$. Fig.1 depicts the block structure of CP-SC and TS-SC. TS is used to mitigate the inter-block-interference(IBE). As long as the duration of TS is longer than that of channel impulse response, IBE effects can be ignored [14]. For the TA-SC scheme, a block of signals $\mathbf{x} = [x_0, x_1, \dots, x_{N-1}]^T$ is transmitted with block N where the last N_p samples are training sequence $\mathbf{s} = [s_0, s_1, \dots, s_{N_p-1}]^T$. Suppose the

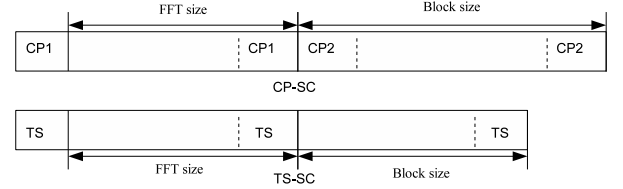


Fig. 1. Block structure of single carrier with cyclic prefix and training sequence

equivalent symbol spaced channel impulse response (CIR) is invariant over each block duration.

$$\mathbf{h}(k) = [h(k, 0), h(k, 1), \dots, h(k, L - 1)]^T; \quad (1)$$

where k denotes the k -th block, L is the channel order and T represents the matrix transpose

At the receiver, we receive the block signal after it passes through the UWA channel modeled linearly by its CIR. The transmitted signal is convolved with the equivalent multipath CIR $h(k)$, which consists of transceiver pulse shaping filter, multipath UWA channel and receiver filter. Additionally, the received signal is subjected to timing offset, Doppler shift and additive white Gaussian noise.

Taking samples at the receiver, the discrete-time signal samples that we can use for detection is given by

$$y(k, n) = \sum_{l=0}^{L-1} h(k, l)x(k, n - l)e^{j2\pi f_k n T_s + \psi_{k,0}} + w(k, n) \quad (2)$$

where T_s is data symbol duration, f_k is Doppler shift in k -th block, caused by relative motion between transceivers, A/D, D/A sampling, variant water current, etc. $\psi_{k,0}$ is phase rotation caused by symbol timing offset in k -th block. $w(k, n)$ is additional white Gaussian noise with power σ^2 . In basic SC-FDE detection, decision is made block-by-block. The first step of detection is to remove the effect of timing offset and Doppler shift. The corresponding k -th received signal block is

$$\mathbf{y}(k) = [y(k, 0), y(k, 1), \dots, y(k, N - 1)]^T; \quad (3)$$

For simplicity and without loss of generality, we eliminate the block index k . For TS-SC transmission, if the length of training sequence is longer than the CIR, the received vector after the remove of timing offset and Doppler shift (include the training sequence) may be written in a matrix form as

$$\mathbf{y} = \tilde{\mathbf{H}}\mathbf{x} + \tilde{\mathbf{w}} \quad (4)$$

where $\tilde{\mathbf{H}}$ represents the effect of the composite channel resulting from transmit filter, UWA physical channel and receiver filter. Thanks to the insertion of training sequence,

the channel matrix $\tilde{\mathbf{H}}$ is circulant with its first column given by the composite CIR(appendd by zeros). Hence, this matrix has the following circulate Toeplitz matrix with the following decomposition:

$$\tilde{\mathbf{H}} = \mathbf{F}^h \mathbf{\Lambda} \mathbf{F} \quad (5)$$

where, matrix $\mathbf{\Lambda} = \tilde{\mathbf{H}} \mathbf{F} \mathbf{F}^h$ is a diagonal matrix formed with the values of the DFT of the composite CIR. Its (k, k) th entry as H_k is given as

$$H_k = \sum_{l=0}^{L-1} h(l) \exp^{-j(2\pi lk/N)} \quad (6)$$

The frequency domain signals of $\mathbf{x}$, and $\mathbf{y}$ is denoted $\mathbf{X} \triangleq \mathbf{F}\mathbf{x}$ and $\mathbf{Y} \triangleq \mathbf{F}\mathbf{y}$, where $\mathbf{F}$ is the normalized DFT matrix of size $N \times N$ and $F_{l,k} = \frac{1}{\sqrt{N}} \exp^{-j(2\pi lk/N)}$, $0 \leq l, k \leq N-1$.

$$\mathbf{Y} = \mathbf{\Lambda} \mathbf{F} \mathbf{x} + \mathbf{F} \mathbf{w} \quad (7)$$

Frequency domain equalization can be expressed as

$$z_1 = \mathbf{F}^h \mathbf{C} \mathbf{\Lambda} \mathbf{F} \mathbf{x} + \mathbf{F}^h \mathbf{C} \mathbf{F} \mathbf{w} \quad (8)$$

, where $\mathbf{C}$ is an $N \times N$ diagonal matrix with its k th diagonal element as the frequency domain equalizer taps $C_k, k = 0, \dots, N-1$. When frequency domain MMSE equalization is considered, the equalizer taps can be expressed as

$$C_k = \frac{H_k^*}{|H_k|^2 + N_0/(2E_s)} \quad (9)$$

then, the k th block symbol of $\hat{\mathbf{x}}_1$ can be approximated as

$$\hat{\mathbf{x}} = \text{dec}(z_1) \quad (10)$$

In summary, the receiver consists of remove of timing offset and Doppler shift, channel estimation and frequency domain equalization. As shown in Fig2, this receiver is a block-by-block decoder.

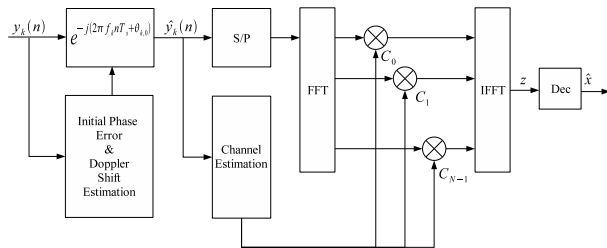


Fig. 2. Receiver structure

III. Receiver algorithm

After AGC and BPF, the received signal is sampled and all processing is performed discretely. If wide-band signal is used, significant Doppler shift will translates each frequency component by a different amount, and the Doppler effect is more accurately modeled as a complete time scaling(expansion or compression) of the signal waveform [18]. In this case, significant Doppler shift is compensated via re-sampling techniques[18] to convert a "wideband" problem into a "narrowband" problem. After re-sampling, initial phase error, residual Doppler rate and channel estimation will be estimated and compensated block by block independently. In this paper, we focus on the processing after re-sampling. In k th block, the phase rotation of n -th symbol can be expressed as

$$\begin{aligned} \psi_{k,n} &= \psi_{k,0} + 2\pi f_k n T_s \\ &= \psi_{k,0} + n \Delta \psi_k, \quad 0 \leq n \leq N-1 \end{aligned} \quad (11)$$

A. Initial average phase error estimation and compensation

The initial average phase error $\psi_{k,0}$ can be obtained from the $(k-1)$ -th training sequence based on ML estimation principles. In AWGN channel, the received training sequence of k -th block can be expressed as

$$\mathbf{Y}_k = \mathbf{S} e^{j\psi_{k,0}} + \mathbf{W} \quad (12)$$

where

$$\mathbf{Y}_k = \begin{bmatrix} y_k(0) \\ \vdots \\ y_k(N_q) \end{bmatrix}, \mathbf{S} = \begin{bmatrix} s(0) \\ \vdots \\ s(N_q) \end{bmatrix}, \mathbf{W} = \begin{bmatrix} w(0) \\ \vdots \\ w(N_q) \end{bmatrix} \quad (13)$$

where $N_q = N_p - 1$. Then, at $k+1$ th block, the ML estimator of initial phase error select $\psi_{k,0}$, which maximizes the likelihood probability

$$\begin{aligned} \Lambda(\psi_{k,0}) &= \log p(y_k(n) | \psi_{k,0}) \\ &= -\frac{1}{N_0} \sum_{n=0}^{N_p-1} |y_k(n) - s(n) e^{-j\psi_{k,0}}|^2 \end{aligned} \quad (14)$$

which is equivalent to selecting that $\psi_{k,0}$ that maximizes

$$\hat{\Lambda}(\psi_{k,0}) = \Re \left\{ e^{-j\psi_{k,0}} \sum_{n=0}^{N_p-1} s^*(n) y_k(n) \right\} \quad (15)$$

Thus, the ML estimator for initial rotating phase is

$$\hat{\psi}_{k,0} = \angle \left\{ \sum_{n=0}^{N_p-1} s^*(n) y_k(n) \right\} \quad (16)$$

where $\angle(\cdot)$ and $(\cdot)^*$ are an phase extracting operation and complex conjugate.

B. Doppler shift estimation and compensation

Two successive training sequences are used to estimate Doppler rate based on ML estimation principles. In AWGN channel, the received training sequence of $k+1$ th block can be expressed as

$$\mathbf{Y}_{k+1} = \mathbf{S}e^{j\psi_{k+1,0}} + \mathbf{W}_{k+1} \quad (17)$$

where $\psi_{k+1,0}$ can be expressed as

$$\psi_{k+1,0} = \psi_{k,0} + N\Delta\psi_k \quad (18)$$

thus,

$$\Delta\psi_k = \frac{\psi_{k+1,0} - \psi_{k,0}}{N} \quad (19)$$

The ML estimation for $\Delta\psi_k$ can be expressed as

$$\hat{\Delta\psi}_k = \angle \left(\sum_{n=0}^{N_p-1} y_k^*(n)y_{k+1}(n) \right) \quad (20)$$

When some prior information about $\Delta\psi_k > 0$ or $\Delta\psi_k < 0$ is available, some modification may be needed on(20). Suppose $\Delta\psi_k > 0$, the modification can be expressed as

$$\tilde{\Delta\psi}_k = \begin{cases} \hat{\Delta\psi}_k & \hat{\Delta\psi}_k > 0 \\ \hat{\Delta\psi}_k + 2\pi & \hat{\Delta\psi}_k < 0 \end{cases} \quad (21)$$

Now, the Doppler shift of k th block can be expressed as

$$\hat{f}_k = \frac{\tilde{\Delta\psi}_k}{2\pi T_s} \quad (22)$$

then, using (11), we can get the value of phase shift $\psi_{k,n}$ as

$$\psi_{k,n} = \psi_{k,0} + n\tilde{\Delta\psi}_k, \quad 0 \leq n \leq N-1 \quad (23)$$

After rotating phase compensation, the k -th received signal can be expressed as

$$\hat{y}_k(n) = y_k(n)e^{-j\psi_{k,n}}, \quad 0 \leq n \leq N-1 \quad (24)$$

After phase rotation compensation, UWA channel is estimated by using training sequence. We assume that the channel is constant within each block. Based on (9), the MMSE equalize coefficient is obtained as $\hat{C}_k, 0 \leq k \leq N-1$. Then

$$\hat{x}_k = \text{dec}[\mathbf{F}^h(\hat{C}_k \mathbf{Y}_k)] \quad (25)$$

IV. Experimental results

Experiments were conducted to verify the proposed TS-SC method in practical ocean environments of South China Sea on June 2007. The bandwidth of the TS-SC signal is $B = 500$ Hz, and the symbol duration is $T_s = 2$ ms. The training sequence is Barker code with its length $N_p = 13$, resulting a interval of $T_g = 26$ ms. The block size $N =$

128 and the useful data size is $(N - N_p) = 115$. The duration of one block is $T = 256$ ms. 65 blocks were transmitted during that experiment. The total number of information bits are 8333 bits and 16666 bits for BPSK and QPSK respectively. The useful data rate are about 450 bps and 900 bps for BPSK and QPSK respectively. Limited by experimental condition, only one receive hydrophones was deployed near the bottom of the ocean with a depth of 40 m. The distance between transmitter and receiver is 30 km. Initial phase error and Doppler shift are estimated block by block. According to the effect of Doppler shift, initial phase error almost increase linearly.

Fig.3 shows the experimental results for TS-SC with BPSK modulation for $N=128$ and $N_p=13$. The initial rotating phase estimation is shown in upper left of Fig.3. The initial rotating phase is estimated block by block. Due to the effect of Doppler shift, the estimated initial rotating phase varies almost linearly. The Doppler shift is estimated by using the successive training sequence block by block. The Doppler shift according to block index is shown in upper right of Fig.3. The average Doppler shift is 0.5104 Hz. The scatter plots in lower left and right of Fig.3 are the input signals and the output signals of FDE respectively. There are 4 error bits in the uncoded systems with a totally 8333 bits.

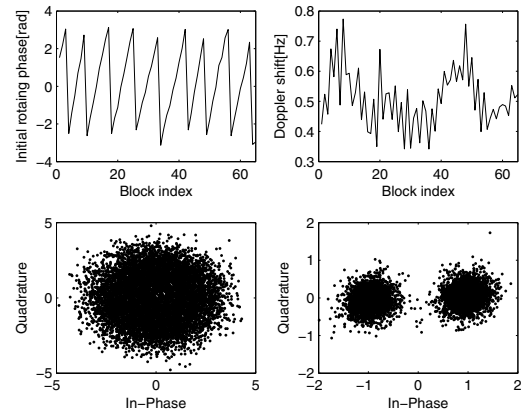


Fig. 3. Experimental results for TS-SC with BPSK modulation at the range of 30km

Fig.4 shows the experimental results for TS-SC with QPSK modulation for $N=128$ and $N_p=13$. The initial rotating phase estimation is shown in upper left of Fig.4. The Doppler shift according to block index is shown in upper right of Fig.4. The average Doppler shift is 0.513 Hz. The scatter plots in lower left and right of Fig.4 are the input signals and the output signals of FDE respectively. There are 103 error bits in the uncoded systems with a totally 16666 bits. In summary, the proposed UWA

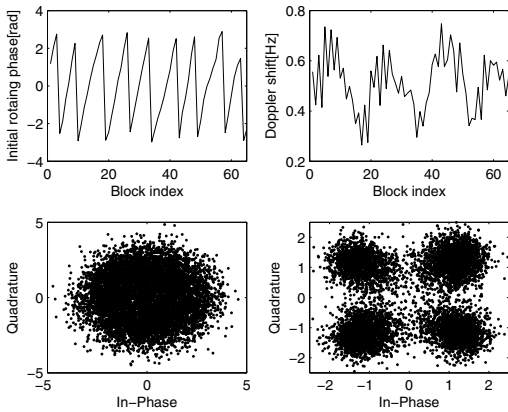


Fig. 4. Experimental results for TS-SC with QPSK modulation at the range of 30km

communication systems embedded with block processing for initial rotating phase, Doppler and channel estimation works well in long range shallow water channel. Limited by experimental condition, only SISO system is considered in this paper. For more practical UAC systems, multichannel receivers and forward error control coding will be employed to improve the system performance.

V. Conclusion

In this paper, training sequence based single carrier (TS-SC) modulation with low frequency frequency domain equalization has been proposed for long range underwater acoustic communication. The maximum likelihood estimators for robust initial phase error, Doppler shift by using training sequence are derived. The receiver parameters are estimated block by block which make it suitable for fast varying channel. The phase rotating caused by initial phase error and Doppler shift is compensated before frequency domain equalization to make the channel estimation more accurate. The experimental results in practical ocean environment shows the achieved good performance of the proposed low complexity UWA communication systems. The proposed method can be easily extended to multichannel receiver.

Acknowledgment

This work was supported in part by National Science Foundation of China (Grant No. 60572098), the Research Fund for the Doctoral Program of Higher Education of China (Grant No. 20070699020), the Doctorate Foundation of Northwestern Polytechnical University (Grant

No. CX200602) and the NPU Foundation for Fundamental Research (Grant No. NPU-FFR-W018102). The author thanks Prof. Haiyan Wang of College of Marine Engineering, Northwestern Polytechnical University for his help.

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